Mini Project - Feature Requirement Document

Overview

This document outlines the detailed requirements for developing the user signup, login, profile management, pricing plan management, and admin panel processes for a web application. The backend should store data securely and manage user sessions, profile data, payments, and access control. It also includes a cron job for plan expiration. Each feature is described in terms of its functionality, expected behavior, acceptance criteria, and how it integrates with other features.

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1. Signup Process

1.1 Social Signup (Account Merge)

- **Requirement**: Implement the ability for users to sign up using their social media accounts (e.g., Facebook, Google).
- **Functionality**: If an account with the same email already exists, the system should merge the social media account with the existing account.
- **Expected Behavior**: Users should be able to seamlessly create an account or merge with an existing one using their social media credentials.
- **Acceptance Criteria**:
    - Users can sign up using at least two social media platforms.
    - If an email already exists, the accounts are merged without data loss.
    - Users are redirected to the dashboard after successful signup.

1.2 Standard Signup

- **Requirement**: Allow users to sign up by providing their email address and creating a password.
- **Functionality**: Upon successful signup, the system should send a confirmation email to the user.
- **Expected Behavior**: Users receive a confirmation email and must confirm their email address to activate their account.
- **Acceptance Criteria**:
    - Users can sign up with an email and password.
    - Confirmation email is sent to the registered email address.
    - Users can confirm their email and activate their account by clicking the link in the email.

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2. Account Verification

2.1 Email Verification through Link

- **Requirement**: Implement email verification through a verification link.
- **Functionality**: After signup, send a verification link to the user's registered email address. The user must click this link to verify their email.
- **Expected Behavior**: The system validates the link and marks the email as verified.
- **Acceptance Criteria**:
    - A verification link is sent to the registered email after signup.
    - Users can click the verification link to verify their email address.
    - The system confirms the email verification upon clicking the link.

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3. Login

3.1 Login

- **Requirement**: Allow verified users to log in using their email and password or through social media accounts.
- **Functionality**: Implement authentication mechanisms for standard and social media logins.
- **Expected Behavior**: Users can log in successfully if their credentials are correct and their account is verified.
- **Acceptance Criteria**:
    - Users can log in using email and password.
    - Users can log in using social media accounts.
    - Only verified accounts can access the dashboard after login.

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4. Profile Management

4.1 Profile Picture Upload (Storj Integration)

- **Requirement**: Provide users with the ability to upload a profile picture, which is stored on Storj.
- **Functionality**: Store the uploaded profile picture on Storj and display it as a thumbnail in the user's profile.
- **Expected Behavior**: Users can see their uploaded profile picture in their profile and on the user listing page.
- **Acceptance Criteria**:
    - Users can upload a profile picture.
    - The profile picture is stored on Storj and displayed as a thumbnail in the user's profile.
    - The profile picture is visible on the user listing page.

4.2 Profile Management (Address, Auto Suggestion, Display in Map)

- **Requirement**: Allow users to manage their profile details, including address information.
- **Functionality**: Implement auto-suggestions for the address field and display the location on a map using Google Maps API.
- **Expected Behavior**: Users can update their address details, receive auto-suggestions, and see their location on a map.
- **Acceptance Criteria**:
    - Users can update their address details.
    - Address field provides auto-suggestions as the user types.
    - The address is displayed on a map within the profile.

4.3 Profile Download

- **Requirement**: Enable users to download their profile information in a specified format.
- **Functionality**: Provide an option for users to export their profile data.
- **Expected Behavior**: Users can download their profile information in a well-formatted document.
- **Acceptance Criteria**:
    - Users can request to download their profile information.
    - Profile data is exported in a user-friendly format.
    - Downloaded profile includes all relevant user details.

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5. Payment Processing and Pricing Plans

5.1 Payment Gateway Integration

- **Requirement**: Integrate a payment gateway, such as Stripe, to process payments for pricing plans.
- **Functionality**: Handle transactions securely and update the user's access level based on the payment made.
- **Expected Behavior**: Users can make payments, and their account is upgraded to the appropriate access level upon successful payment.
- **Acceptance Criteria**:
    - Users can initiate payments through the payment gateway.
    - Payment transactions are secure and processed correctly.
    - User access level is upgraded immediately upon successful payment.

5.2 Pricing Plans with Time-Limited Access

- **Requirement**: Provide time-limited pricing plans with varying benefits. Plans should expire after a set duration (1 hour, 6 hours, or 12 hours).
- **Functionality**: Upon payment, the selected plan (Free, Silver, Gold) is activated and expires automatically based on its duration.

- **Expected Behavior**: Users can select a pricing plan, and the system automatically disables access when the plan's time expires.
- **Acceptance Criteria**:
    - Users can select and pay for a plan (Free, Silver, Gold).
    - Plans activate immediately upon payment and are stored in the user profile.
    - The system runs a cron job to check plan durations and automatically expire plans after 1 hour, 6 hours, or 12 hours.
    - The plan status is stored in the user's profile and displayed in the admin panel.

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6. Admin Panel

6.1 Admin Role

- **Requirement**: Implement an admin role with special privileges to manage the application.
- **Functionality**: Admins can log in with their credentials to access the admin panel.
- **Expected Behavior**: Admins have access to advanced management features within the application.
- **Acceptance Criteria**:
    - Admins can log in with special credentials.
    - Admin panel is accessible only to admin users.
    - Admins have access to user management and other administrative features.

6.2 User Listing with Search, Filter, and Pagination

- **Requirement**: Provide an interface for admins to view a list of all users.
- **Functionality**: Implement search, filter, and pagination functionalities to help admins manage users effectively.
- **Expected Behavior**: Admins can search for users, apply filters, and navigate through pages of user listings.
- **Acceptance Criteria**:
    - Admins can view all users in a list.
    - Admins can search and filter users based on various criteria.
    - User list is paginated for easy navigation.

6.3 User Listing

- **Requirement**: Display all users in a list format.
- **Functionality**: Show basic information of each user in the list.
- **Expected Behavior**: Admins can see a comprehensive list of users with basic details.
- **Acceptance Criteria**:
    - All users are displayed in a list.
    - Basic information such as name, email, and status is shown for each user.

6.4 User Listing Search

- **Requirement**: Enable admins to search for specific users using various criteria.
- **Functionality**: Implement search functionality based on name, email, or other relevant fields.
- **Expected Behavior**: Admins can quickly find specific users using the search feature.
- **Acceptance Criteria**:
    - Admins can search users by name, email, or other criteria.
    - Search results are accurate and displayed quickly.

6.5 User Listing with Pagination

- **Requirement**: Implement pagination for the user listing to manage large numbers of users efficiently.
- **Functionality**: Allow admins to navigate through different pages of the user list.
- **Expected Behavior**: Admins can easily browse through user information in a paginated format.
- **Acceptance Criteria**:
    - User list is divided into pages.
    - Admins can navigate between pages to view more users.
    - Pagination controls are user-friendly and responsive.

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7. Cron Job for Plan Expiration

7.1 Plan Expiration Cron Job

- **Requirement**: Implement a cron job to check for expired plans based on their duration (1 hour, 6 hours, 12 hours) and deactivate them accordingly.
- **Functionality**: The cron job should run periodically to validate plan expiration times and revoke access to expired plans.
- **Expected Behavior**: Plans expire at the end of their duration, and users lose access to premium features if the plan expires.
- **Acceptance Criteria**:
    - The cron job runs at regular intervals to check for plan expirations.
    - Plans are deactivated automatically once their duration (1 hour, 6 hours, or 12 hours) ends.
    - Expired plans update the user’s profile and reflect in the admin panel.

FAQ

Q: Which social login options should we implement?
A: You can choose your own social login options, but Facebook and Google are recommended as they are easier to implement than others.

Q: Which payment gateway should we use?
A: Stripe's free/sandbox version is recommended for handling payments.

Q: Which service should be used for address updates and suggestions?
A: Google Maps API should be used for auto-suggestions and map pin display for address updates.

Q: Where should user files, like profile pictures, be stored?
A: Storj is recommended for storing user files securely.

Q: Are there any special offerings on the pricing plans?
A: There are no special offerings. Users can pay for the plan of their choice, and plan details must be stored and linked with the user.

Q: How should user search functionality be handled in the admin panel?
A: User search by name should be managed with filters and pagination. Refreshing the page should not reset search criteria or pagination.

Q: How long do the pricing plans last, and how is this managed?
A: Pricing plans have time limits of 1 hour, 6 hours, or 12 hours. A cron job will manage plan expiration and automatically revoke access when a plan's duration ends.